

MyTZStudies.com

Your Free Tanzania Exam Library

Past papers, answer keys, and study resources
for Tanzanian students.

MyTZStudies.com

Past Exam Papers	Worked Answer Keys	Free Access
Choose Your Level	Study Step by Step	Learn at Your Pace

Standards 1-7 | Form 2 | Form 4

Share this with a friend.
Every student deserves free study materials.

MATHEMATICS

STANDARD FIVE

Terminal exam 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	02
Level	Standard Five
Year	2024
Exam Board / Issuer	KIPCO
Exam	Terminal exam
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S5-2024-KIPCO-Terminal-exam.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Calculations

1(i). Add $4999 + 443276$.

Step 1: 1 place: $9 + 6 = 15$; write 5, carry 1.

Step 2: 10 place: $9 + 7 + \text{carry } 1 = 17$; write 7, carry 1.

Step 3: 100 place: $9 + 2 + \text{carry } 1 = 12$; write 2, carry 1.

Step 4: 1000 place: $4 + 3 + \text{carry } 1 = 8$; write 8.

Step 5: 10000 place: $0 + 4 = 4$; write 4.

Step 6: 100000 place: $0 + 4 = 4$; write 4.

Step 7: $4999 + 443276 = 448275$.

Final Answer: 448275

1(ii). Subtract 9804 from 11022 .

Step 1: 1 place: regroup 1 from the next place; $12 - 4 = 8$.

Step 2: 10 place: $1 - 0 = 1$.

Step 3: 100 place: regroup 1 from the next place; $10 - 8 = 2$.

Step 4: 1000 place: regroup 1 from the next place; $10 - 9 = 1$.

Step 5: 10000 place: $0 - 0 = 0$.

Step 6: $11022 - 9804 = 1218$; check $1218 + 9804 = 11022$.

Final Answer: 1218

1(iii). Multiply 922×37 .

Step 1: Split into place values: $900 \times 37 = 33300$; $20 \times 37 = 740$; $2 \times 37 = 74$.

Step 2: Add the partial products: 34114.

Final Answer: 34114

1(iv). Add 6573 to 45768.

Step 1: 1 place: $3 + 8 = 11$; write 1, carry 1.

Step 2: 10 place: $7 + 6 + \text{carry } 1 = 14$; write 4, carry 1.

Step 3: 100 place: $5 + 7 + \text{carry } 1 = 13$; write 3, carry 1.

Step 4: 1000 place: $6 + 5 + \text{carry } 1 = 12$; write 2, carry 1.

Step 5: 10000 place: $0 + 4 + \text{carry } 1 = 5$; write 5.

Step 6: $6573 + 45768 = 52341$.

Final Answer: 52341

1(viii). Divide 14148 by 18.

Step 1: $141 \div 18$: write 7; $7 \times 18 = 126$; $141 - 126 = 15$.

Step 2: $154 \div 18$: write 8; $8 \times 18 = 144$; $154 - 144 = 10$.

Step 3: $108 \div 18$: write 6; $6 \times 18 = 108$; $108 - 108 = 0$.

Step 4: $18 \times 786 = 14148$.

Step 5: $14148 \div 18 = 786$.

Final Answer: 786

2(iii). 10000 cows; 7011 die. Alive?

Step 1: 1 place: regroup 1 from the next place; $10 - 1 = 9$.

Step 2: 10 place: regroup 1 from the next place; $9 - 1 = 8$.

Step 3: 100 place: regroup 1 from the next place; $9 - 0 = 9$.

Step 4: 1000 place: regroup 1 from the next place; $9 - 7 = 2$.

Step 5: 10000 place: $0 - 0 = 0$.

Step 6: $10000 - 7011 = 2989$; check $2989 + 7011 = 10000$.

Final Answer: 2989 cows

3(i). Handeni 561217 seedlings, Manyoni 393888. Total?

Step 1: 1 place: $7 + 8 = 15$; write 5, carry 1.

Step 2: 10 place: $1 + 8 + \text{carry } 1 = 10$; write 0, carry 1.

Step 3: 100 place: $2 + 8 + \text{carry } 1 = 11$; write 1, carry 1.

Step 4: 1000 place: $1 + 3 + \text{carry } 1 = 5$; write 5.

Step 5: 10000 place: $6 + 9 = 15$; write 5, carry 1.

Step 6: 100000 place: $5 + 3 + \text{carry } 1 = 9$; write 9.

Step 7: $561217 + 393888 = 955105$.

Final Answer: 955105 seedlings

5(ii). Halima 34656 bags, sells half. Remaining?

Step 1: $3 / 2$: write 1; $1 \times 2 = 2$; $3 - 2 = 1$.

Step 2: $14 / 2$: write 7; $7 \times 2 = 14$; $14 - 14 = 0$.

Step 3: $6 / 2$: write 3; $3 \times 2 = 6$; $6 - 6 = 0$.

Step 4: $5 / 2$: write 2; $2 \times 2 = 4$; $5 - 4 = 1$.

Step 5: $16 / 2$: write 8; $8 \times 2 = 16$; $16 - 16 = 0$.

Step 6: $2 \times 17328 = 34656$.

Step 7: $34656 / 2 = 17328$.

Final Answer: 17328 bags

6(i). 50 matchsticks per box;167 boxes. Total?

Step 1: Split into place values: $50 \times 167 = 8350$.

Step 2: Add the partial products: 8350.

Final Answer: 8350 sticks

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(ix). Add $1.09 + 11.93$.

Step 1: Hundredths $9 + 3 = 12$; write 2 carry 1.

Step 2: Tenths $0 + 9 + 1 = 10$; write 0 carry 1.

Step 3: Whole parts $1 + 11 + 1 = 13$.

Final Answer: 13.02

1(v). Calculate $5/9 + 3/7$.

Step 1: Common denominator 63: $5/9 = 35/63$; $3/7 = 27/63$.

Step 2: $35 + 27 = 62$.

Final Answer: 62/63

1(vi). Subtract $1/5 - 1/8$.

Step 1: Common denominator 40: $1/5 = 8/40$; $1/8 = 5/40$.

Step 2: $8 - 5 = 3$.

Final Answer: 3/40

1(vii). Multiply $7/12 \times 11/24$.

Step 1: Numerators $7 \times 11 = 77$.

Step 2: Denominators $12 \times 24 = 288$.

Step 3: No common factor.

Final Answer: 77/288

1(x). Subtract $15 - 10.09$.

Step 1: Write 15.00 and 10.09 aligned.

Step 2: Borrow: 14 whole, 9 tenths, 10 hundredths.

Step 3: Hundredths $10 - 9 = 1$; tenths $9 - 0 = 9$; whole $14 - 10 = 4$.

Final Answer: 4.91

2(i). Write three hundred thousand eighty-one thousand and three goats in figures.

Step 1: Combine $300000 + 81000 + 3$.

Step 2: $300000 + 81000 + 3 = 381003$.

Final Answer: 381003

2(ii). Find sum of two consecutive numbers before 18041.

Step 1: Numbers 18039 and 18040.

Step 2: $18039 + 18040 = 36079$.

Final Answer: 36079

2(iv). Write 10.5 in words.

Step 1: Read 10 then decimal digit 5.

Final Answer: Ten point five

2(v). Total value of the ten-thousands digit in 345556?

Step 1: Digit 4 in ten-thousands.

Step 2: $4 \times 10000 = 40000$.

Final Answer: 40000

2(vi). Form smallest number using 7,0,6,5,3,4 once each.

Step 1: First digit must be nonzero; choose 3.

Step 2: Remaining ascending 0,4,5,6,7.

Final Answer: 304567

3(ii). Three brothers have successive age differences 12 years; middle is 21. Youngest age?

Step 1: Youngest is 12 years below middle.

Step 2: $21 - 12 = 9$.

Final Answer: 9 years

3(iii). Write 213 in Roman numerals.

Step 1: $200=CC$; $10=X$; $3=III$.

Final Answer: CCXIII

4(i). Which number is both even and prime?

Step 1: An even number above 2 has factors 1,2, itself.

Step 2: 2 has only 1 and 2.

Final Answer: 2

4(ii). Change $27/4$ into an improper fraction, as requested.

Step 1: $27 > 4$, so $27/4$ is already improper.

Step 2: If mixed form were intended, $27 / 4$ gives 6 remainder 3.

Final Answer: $27/4$ already improper (mixed form $6 \frac{3}{4}$).

4(iii). Convert $2 \frac{1}{2}$ km to centimetres.

Step 1: $1 \text{ km} = 100000 \text{ cm}$.

Step 2: $2.5 \times 100000 = 250000$.

Final Answer: 250000 cm

5(i). Which number multiplied by itself equals 361?

Step 1: $19 \times 19 = 361$.

Step 2: Also $(-19) \times (-19) = 361$ if negative integers allowed.

Final Answer: 19 (also -19 over all integers)

5(iii). 3 children share 3 crates of soda equally. Bottles per child?

Step 1: $3 / 3 = 1$ crate each.

Step 2: The number of bottles in one crate is not printed.

Final Answer: 1 crate per child; bottle count cannot be determined without crate capacity (24 if a 24-bottle crate is assumed).

6(ii). A car covers 5 km in 3 minutes. Time in seconds for 2 km at same speed?

Step 1: $3 \times 60 = 180$ seconds for 5 km.

Step 2: $180 / 5 = 36$ seconds per km.

Step 3: $36 \times 2 = 72$.

Final Answer: 72 seconds

6(iii). Donny buys 3 plates for 145000 shillings. How many more plates if spending 290000?

Step 1: $290000 / 145000 = 2$ times the cost.

Step 2: $3 \times 2 = 6$ plates in total.

Step 3: $6 - 3 = 3$ more.

Final Answer: 3 more (6 total)

7(i). Flowers pictogram: Monday 6 symbols, Tuesday 4, Wednesday 7; each 17 flowers. Total?

Step 1: $6 + 4 + 7 = 17$ symbols.

Step 2: $17 \times 17 = 289$.

Final Answer: 289 flowers

7(ii). Same pictogram: Monday 6 symbols, Tuesday 4, each 17 flowers. Total both days?

Step 1: $6 + 4 = 10$ symbols.

Step 2: $10 \times 17 = 170$.

Final Answer: 170 flowers

8(i). Draw a parallelogram.

Step 1: Draw two parallel horizontal sides.

Step 2: Join ends with two parallel sloping sides.

Final Answer: Four-sided figure with both pairs of opposite sides parallel.

8(ii). Find perimeter of isosceles triangle with equal sides 14 cm, base 24 cm; printed height 12 cm.

Step 1: Tick marks make both sloping sides 14 cm.

Step 2: $14 + 14 + 24 = 52$.

Step 3: Height 12 does not lie on the boundary; it is geometrically inconsistent with side 14 and half-base 12.

Final Answer: 52 cm from boundary labels

8(iii). Buy 1 kg maize at 2000 sh/kg, 2 kg beans at 2500 sh/kg, 2 soap bars at 1500 sh each. Total?

Step 1: Maize $1 \times 2000 = 2000$.

Step 2: Beans $2 \times 2500 = 5000$.

Step 3: Soap $2 \times 1500 = 3000$.

Step 4: $2000 + 5000 + 3000 = 10000$.

Final Answer: Sh 10000

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Source Checks

Darasa Huru - KIPCO Standard Five Terminal Examinations 2024